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RESEARCH

CNC toolpath-driven thermal modeling of WAAM: thermal assessment and microstructural interpretation

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Abstract
Wire Arc Additive Manufacturing (WAAM) involves highly transient thermal cycles, repeated reheating, and strong spatial temperature gradients, all of which govern the final microstructure of the deposited material. In this work, a three-dimensional transient thermal model for WAAM of ER70S-6 carbon steel was developed directly from the same processed CNC G-code used to operate the experimental system. This approach allowed the simulation to preserve the real deposition trajectory, torch motion, arc on/off sequence, and fan-assisted interpass cooling intervals. The computational workflow combined G-code parsing, geometric reconstruction of the deposited wall and substrate, structured voxelization, progressive material activation, and explicit transient heat-transfer solution. The thermal problem was formulated considering conduction, convection, radiation, and a moving volumetric Gaussian heat source following the deposition path. As a representative case study, the framework was applied to an eight-bead thin-wall deposition produced under a single controlled GMAW condition and compared with thermographic measurements acquired during processing. The simulated temperature fields showed good qualitative agreement with infrared observations during both active deposition and interpass cooling. Quantitative comparisons of maximum surface temperature and local temperature histories also showed that the model reproduced the main thermal evolution of the process, despite the saturation and spatial-resolution limitations of the thermal camera. The simulated thermal histories were interpreted alongside spatially corresponding metallographic observations along the build height. Faster cooling near the substrate was associated with finer acicular ferrite, repeated reheating in the intermediate region promoted more homogeneous equiaxed polygonal ferrite, and slower cooling in the upper region preserved a coarser as-deposited ferritic morphology.

Keywords Wire Arc Additive Manufacturing (WAAM) · Transient thermal modeling · G-code-driven simulation · Voxel-based modeling · Microstructural evolution · Interlayer thermal cycling

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1 Introduction

Wire Arc Additive Manufacturing (WAAM) has emerged as one of the most promising metal additive manufacturing technologies to produce medium- and large-scale components due to its high deposition rate, relatively low equipment cost, and efficient material usage [1–4]. Unlike powder-based processes, WAAM relies on the melting of wire feedstock through an electric arc, which makes it especially attractive for industries already familiar with welding technologies, robotic deposition, and large-scale fabrication [1–4]. In addition, because the process is not constrained by a powder bed or closed chamber, WAAM offers important advantages for the fabrication of large structural parts in sectors such as aerospace, energy, and construction [1, 4, 5].

The WAAM process is characterized by complex and highly transient thermal interactions, including repeated heating and cooling cycles, localized melting and resolidification, steep spatial temperature gradients, and strong thermal coupling between the deposited material and the substrate [6–8]. These thermal conditions significantly affect bead geometry, residual stress distribution, microstructural evolution, grain growth, and ultimately the mechanical performance of the part [6–9]. In particular, the progressive transition from strong heat extraction by the substrate to a more thermally insulated deposition over previously deposited material promotes heat accumulation, modifies interpass temperature, and changes local cooling conditions along the build height [6, 7]. As a result, the final structure of WAAM components becomes strongly position-dependent, both thermally and metallurgically.

Transient thermal simulation is therefore central to the analysis of WAAM because it provides access to thermal variables that cannot be fully resolved from nominal process parameters alone, including local peak temperature, interpass temperature, cooling rate, reheating frequency, and thermal gradients [6, 10–13]. These variables control the thermal path followed by each region of the deposit and directly influence solidification, austenitization, phase transformation, grain refinement, grain coarsening, and microstructural heterogeneity along the build height [8–10, 12]. For carbon steels processed by WAAM, rapid cooling near the substrate can favor finer ferritic products, whereas repeated reheating and slower cooling in upper or intermediate regions can promote grain growth, microstructural homogenization, or the development of coarser ferritic morphologies [8–10]. Therefore, the final microstructure cannot be interpreted solely from voltage, current, travel speed, or wire feed speed, but must be related to the local thermal history experienced at each position of the component.

A particularly relevant challenge in many thermal simulations is the lack of direct correspondence between the model input and the actual process-execution sequence [6, 11, 13–15]. In many studies, simplified paths, idealized heat inputs, or assumed deposition sequences are used, which may differ from the instructions ultimately executed by the manufacturing system [11, 13, 14]. This can lead to a loss of process-execution consistency when controller-programmed events such as non-deposition movements, arc interruptions, interlayer dwell periods, and auxiliary cooling intervals are not represented in the thermal model. Consequently, the temporal evolution of heat input and cooling conditions may differ from that of the physical process, affecting the reconstructed cooling rates, reheating intensity, and local thermal gradients relevant to process–structure interpretation [16]. The present approach addresses this issue by reconstructing these time-ordered process states from the processed machine instructions; it does not, however, constitute a full representation of controller-level motion dynamics or manipulator-specific kinematics.

Recent process-aware simulation approaches have already established several of the numerical ingredients employed in the present work. Zhang et al. [11] introduced a scalable process-aware thermal framework based on path-level discretization and contact-graph thermal interactions, while Mashhood et al. [15] demonstrated the translation of AM G-code into a thermomechanical finite-element activation sequence for selective laser melting. In the WAAM context, Denkena et al. [14] developed a Dexel-based multilayer approach for efficient geometry prediction and digital-twin integration. Progressive material activation and moving heat-source formulations are also well established in additive-manufacturing thermal simulation. Therefore, G-code-driven activation, voxel-based representation, and Gaussian heat-source modeling are not considered individual methodological novelties of the present work.

The contribution of the present framework lies instead in the process-level integration of these established numerical components using the same post-processed CNC instruction file employed during physical WAAM fabrication. Rather than extracting only the deposition trajectory, the framework reconstructs the time-resolved machine-execution sequence, including torch motion, deposition and non-deposition states, arc activation and extinction, programmed interpass dwell periods, and auxiliary fan operation. These signals are used synchronously to update progressive material activation, the moving volumetric heat source, and time-dependent anisotropic convective boundary conditions. In addition, the activated deposition geometry is informed by experimentally measured bead dimensions rather than only by the nominal toolpath geometry. The resulting contribution is therefore not a new thermal governing equation or activation algorithm, but a process-consistent computational architecture that preserves the chronological coupling

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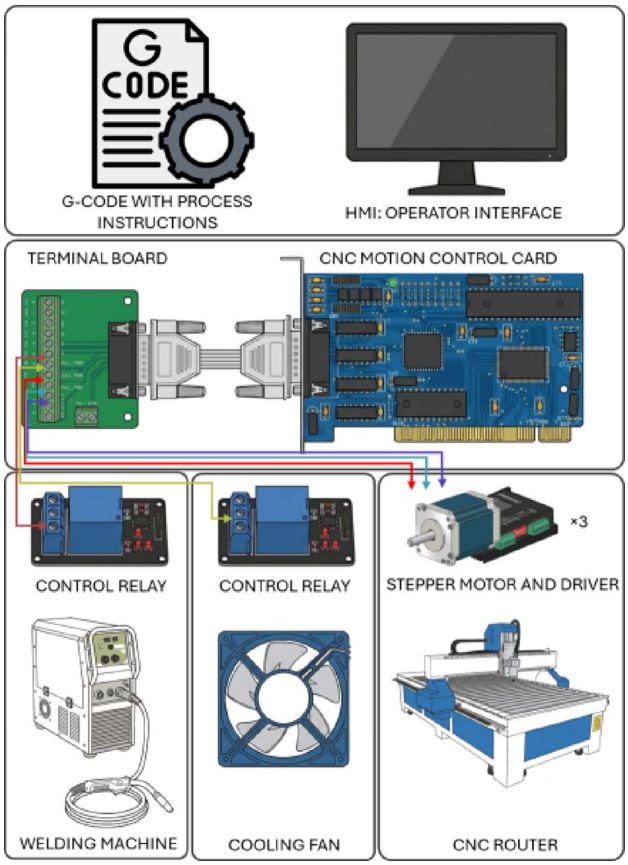


Fig. 1 Integrated WAAM system showing G-code-driven control of CNC motion, welding arc activation, and auxiliary cooling

between machine execution, material addition, heat input, and controlled cooling in CNC-based WAAM. This enables the reconstructed local thermal histories to be directly compared with thermographic measurements and spatially interpreted alongside the experimentally observed microstructural variation.

2 Materials and methods

2.1 WAAM experimental setup and toolpath generation

The WAAM system consists of a Gas Metal Arc Welding (GMAW) power source integrated with a three-axis CNC router adapted for additive manufacturing. The welding torch is mounted on the CNC motion system, enabling controlled deposition along programmed trajectories. The motion platform is controlled through NCStudio software, which interprets G-code commands and coordinates the movement of the machine axes. In addition to motion control, peripheral signals originally intended for spindle operation are repurposed to control key WAAM functions, including arc activation and the operation of an auxiliary cooling fan between deposition passes [17].

As shown in Fig. 1, the system integrates multiple hardware and control components, including the CNC motion control card, terminal board, stepper motor drivers, relay modules, and external devices such as the welding machine and cooling fan. The G-code instructions are first interpreted at the control level and then translated into coordinated motion commands and electrical signals that activate both the deposition process and auxiliary systems.

Toolpaths were initially generated using Ultimaker Cura slicing software, where the deposition strategy was defined in terms of layer-by-layer trajectories, bead spacing, layer height, and travel speed. Since the resulting G-code was not directly compatible with the WAAM system, a self-developed Python post-processing script was used to generate the final NC file. This script adapted the toolpath to the specific control logic of the CNC-based WAAM setup by incorporating arc ignition and extinction commands, travel-speed settings, interpass cooling intervals, and fan activation signals. Therefore, the final post-processed G-code contained both the geometric trajectory and the time-resolved process commands required for machine execution and thermal simulation.

2.2 Raw materials

The deposition material employed in this work was ER70S-6 carbon steel wire with a nominal diameter of 0.9 mm. ASTM A36 steel plates measuring 150 × 25 × 10 mm were used as the base material. The selection of both materials was based on their comparable chemical composition, which facilitates metallurgical compatibility and reduces compositional discontinuities during the initial deposition stages. The elemental composition of both the filler wire and the substrate was determined by Optical Emission Spectrometry (OES) using a Q8 Magellan spectrometer and is summarized in Table 1. All depositions were performed using a shielding gas composed of 95% Argon (Ar) and 5% Carbon Dioxide (CO_2), supplied at a constant flow rate of 16.5 L/min. This Ar-rich mixture with a small active CO_2 addition was selected because shielding-gas composition strongly affects arc stability, metal transfer behavior, bead geometry, and oxidation in GMAW-based WAAM. Low CO_2 additions

Table 1 Chemical composition of the feedstock materials

Material	C [%]	Mn [%]	Si [%]	P [%]	S [%]	Cu [%]	Cr [%]	Fe [%]
Filler	0.08	1.52	0.88	0.012	0.012	0.19	0.026	Balance
Substrate	0.159	1.220	0.232	0.020	0.004	0.020	0.018	Balance

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provide an active component that supports arc stability and wetting while limiting excessive oxidation, spatter, droplet coarsening, and transfer instability commonly associated with higher CO_2 fractions [18].

2.3 Deposition conditions and reference thin-wall fabrication

The WAAM process was conducted using a GMAW power source (SKY MIG 4060 TX, SWEISS) operating in short-circuit transfer mode under constant-voltage conditions. For experimental thermal assessment and extraction of representative geometric and electrical parameters, a single deposition condition was selected, consisting of an arc voltage of 20 V, a wire feed speed of 6 m/min, a travel speed of 250 mm/min, and an electrode stick-out of approximately 10 mm. Under these conditions, an eight-bead thin-wall structure was fabricated, with each deposited layer corresponding to a single bead. This geometry provided a controlled and repeatable configuration for experimental characterization, numerical comparison, and direct analysis of interlayer thermal behavior and bead geometry.

To limit excessive heat accumulation, a controlled dwell time was introduced between successive layers, allowing partial cooling of the deposited material. This cooling stage was further assisted by forced air convection using an auxiliary fan, contributing to more stable interpass thermal conditions.

Process monitoring was carried out using a weld data acquisition system (CWT WDL-3), which continuously recorded arc voltage, current, and wire feed speed during deposition. The data were collected through a serial interface and processed using custom Python scripts, enabling verification of process stability and providing input parameters for the thermal model.

Figure 2 shows the three-dimensional geometry of the fabricated thin wall, together with the corresponding deposition trajectories generated in the slicing software (Cura), which were subsequently post-processed and used for both machine execution and thermal simulation.

2.4 Metallographic preparation and microstructural characterization

Microstructural characterization was carried out by optical metallography following standard specimen preparation procedures in accordance with ASTM E3. After deposition, the thin-wall sample was sectioned in the transverse direction, perpendicular to the deposition path, mounted in resin, ground with SiC abrasive papers, polished, and chemically etched with Nital solution to reveal the steel microstructure. The analysis was performed on the same representative wall used to define the geometric parameters of the thermal model, allowing the observed microstructural variation to be spatially associated with the corresponding simulated thermal histories.

To evaluate microstructural variation from the substrate region to the top of the deposit, a continuous reconstruction of the transverse section was generated from optical micrographs acquired at 500× magnification using an optical microscope. The reconstruction covered a central vertical strip of the wall cross-section, consisting of 2 micrographs in width and 70 micrographs in height, resulting in a continuous mosaic of 140 stitched images. This central strip was selected to sample the build height while avoiding lateral edge effects.

In addition to qualitative metallographic examination, the reconstructed mosaic was processed using an automated image-analysis routine to quantify grain-size variations along the build direction. The segmentation pipeline, including its validation against manual and semi-automatic grain-size measurements and reference values reported in the literature, was described in detail in a previous work by the authors [19] and was adopted here without modification. Segmentation quality was verified by visual inspection of detected grain boundaries against the underlying reconstructed micrograph. Grains touching the outer frame of the reconstructed field were excluded to avoid geometric distortion, following the same criterion adopted in [19].

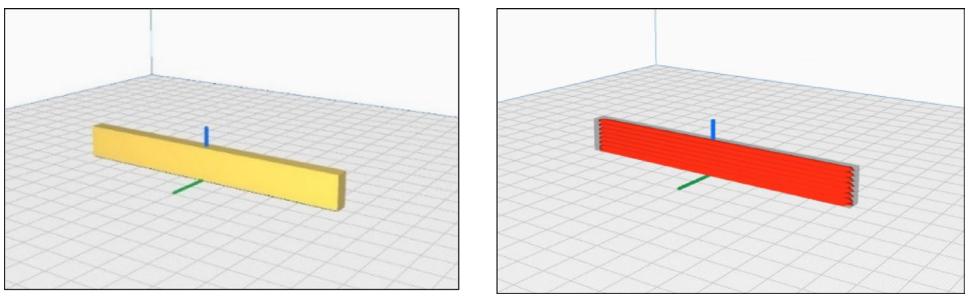


Fig. 2 Fabricated thin-wall geometry and associated deposition trajectories from slicing software

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2.5 Infrared thermography and thermal data acquisition

The thermal evolution during deposition and interpass cooling was monitored with a Fluke Ti300 infrared camera (measurement range − 20 to 1200 °C), positioned approximately 1 m from the deposited wall at an oblique angle with both horizontal and vertical components, so that the complete wall, the bead surface, and the surrounding heat-affected region remained within the field of view throughout the process.

The oblique viewing configuration also introduces uncertainty in the pointwise temperature comparison because the projected pixel footprint and apparent surface orientation vary across the observed wall.

Thermographic recordings were processed in Python to isolate the deposited wall, generate temperature-field visualizations, and extract pointwise temperature histories at representative positions along the wall height, which were then compared with the corresponding simulated histories.

An effective emissivity value of 0.70 was adopted throughout the thermal analysis. This value was selected as a representative constant for low-carbon steel at elevated temperatures, since published data report emissivity values close to 0.7 in the 500–800 °C range, while also showing that emissivity depends on temperature, surface condition, and oxidation state [20, 21]. Therefore, $\varepsilon = 0.70$ was used as an effective radiometric parameter for consistent comparison between thermographic measurements and simulations, and as the emissivity value for the radiative boundary condition in the thermal model [21, 22]. Although this simplification does not capture emissivity changes during heating, cooling, and progressive surface oxidation, it provides a consistent basis for the qualitative and quantitative thermal comparisons performed in this study. Together with the finite spatial resolution, oblique viewing configuration, emissivity uncertainty, and upper measurement limit of the camera, these factors restrict the thermographic comparison to a representative assessment of the thermal response rather than a pointwise validation of the complete temperature field.

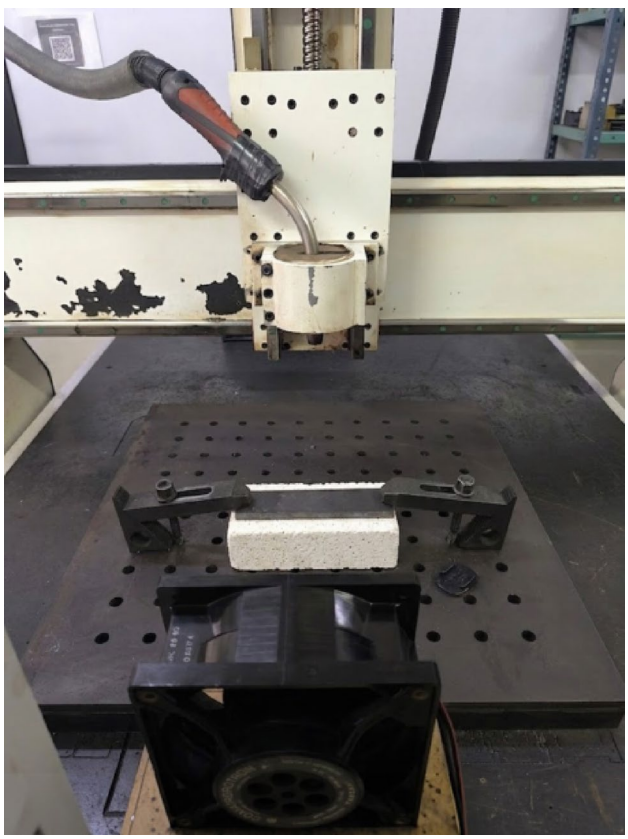


Fig. 3 Experimental WAAM setup showing the CNC-based system, perpendicular forced cooling using an auxiliary fan, and thermal insulation of the substrate using a refractory brick

2.6 Experimental setup and boundary condition control

The experimental configuration used in this study is shown in Fig. 3. The WAAM system was complemented with additional elements designed to control and reproduce specific thermal boundary conditions required for both the experiments and the numerical model. A forced cooling system was implemented using an external fan positioned perpendicular to the deposition direction. This configuration ensures a directional convective effect on one side of the deposited wall, introducing an anisotropic cooling condition that significantly influences heat dissipation and thermal gradients during the process.

To limit heat transfer through the substrate, the base plate was placed over a refractory brick. This configuration substantially reduces heat conduction toward the worktable and supporting structure, allowing the lower substrate surface to be approximated as adiabatic in the thermal model. This boundary condition represents the insulating effect of the refractory support rather than a complete elimination of heat transfer. Residual conductive heat loss through the substrate–brick contact may still occur and could influence the predicted cooling response near the substrate; this contribution is neglected in the present model.

Additionally, the same auxiliary fan used in the experimental setup is incorporated into the process control logic through the G-code, enabling synchronized activation of forced convection during interlayer dwell periods. This ensures that both the physical experiment and the numerical simulation share equivalent cooling conditions.

Figure 3 illustrates the overall configuration, including the CNC-based WAAM system, the positioning of the

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cooling fan, and the insulated substrate arrangement used to control thermal boundary conditions.

2.7 Computational workflow

The proposed methodology follows a sequential workflow in which the post-processed machine instructions are used as the primary input for the thermal simulation. In the present CNC-based implementation, a dedicated parser interprets the NCStudio-compatible G-code together with the specific control logic used to activate the welding arc and auxiliary cooling system. From these instructions, motion commands, deposition and non-deposition segments, arc states, interpass dwell periods, travel-speed information, and fan activation events are extracted and converted into a time-resolved set of process states, as illustrated in Fig. 4. These process states provide the chronology required for geometric reconstruction, progressive material activation, moving heat-source synchronization, and time-dependent boundary-condition assignment.

The instruction-extraction stage is inherently dependent on the machine controller and its programming conventions, whereas the subsequent thermal-simulation workflow is based on the reconstructed process states rather than on a specific controller language. For other manufacturing systems, including six-axis robotic WAAM, the main adaptation would therefore occur at the parsing stage. Robotic toolpaths may originate from a generic CAM or NC representation and subsequently be converted through a postprocessor into controller-specific robot instructions, or they may be defined directly in the robot programming environment. Accordingly, the required process information may be extracted either from the trajectory representation before controller-specific postprocessing or from the resulting robot program. In the latter case, the parser would additionally need to account for robot-specific information such as tool pose and orientation, multi-axis interpolation, and controller timing. Once the tool trajectory, deposition state, arc state, travel speed, dwell periods, and auxiliary-process events are converted into the required process-state representation, the subsequent thermal-modeling architecture can be retained.

For the present CNC implementation, the effective bead width and layer height are then used to geometrically expand each G1 deposition segment and reconstruct the deposited volume, while the substrate is generated according to its known dimensions. The resulting geometry is mapped onto a structured voxel mesh in which deposited voxels are activated progressively in synchrony with the reconstructed torch trajectory and deposition state. Consequently, the computational domain grows layer by layer following the actual deposition chronology, as illustrated in Fig. 5.

The transient heat-transfer problem is subsequently solved over the evolving active domain using the explicit finite-difference formulation implemented in the self-developed Python code. During the simulation, the reconstructed

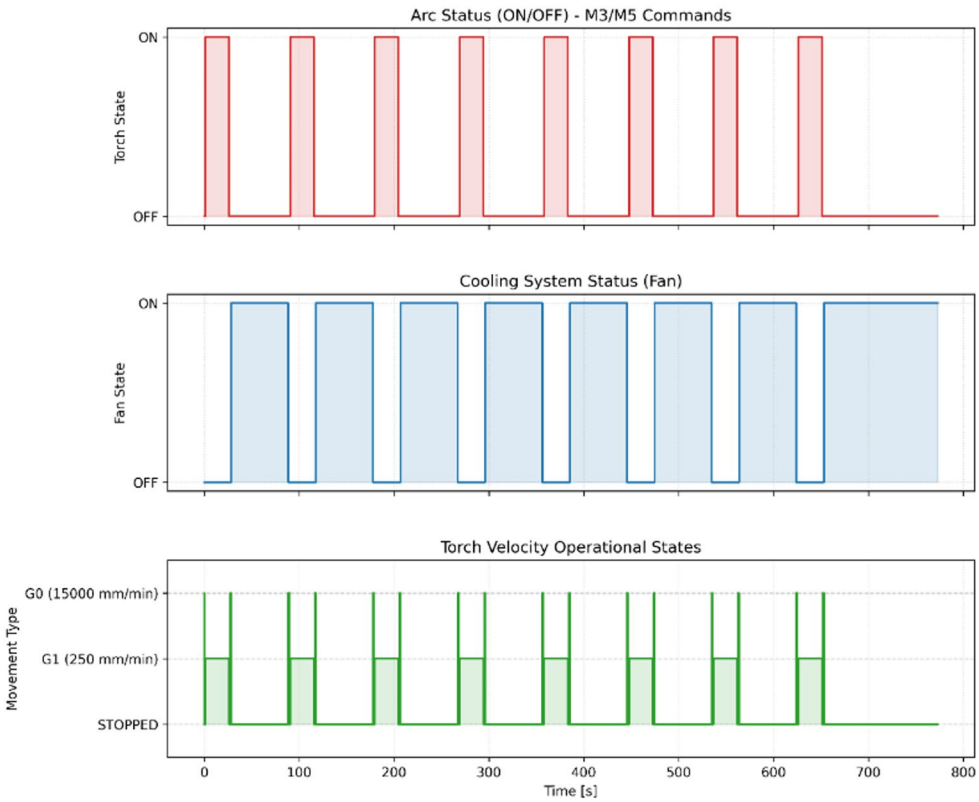


Fig. 4 Time-resolved process states extracted from the machine instructions and used to drive the thermal simulation. For the present CNC implementation, the states are obtained from the post-processed G-code and include arc activation (top), auxiliary cooling operation (middle), and torch-motion state (bottom)

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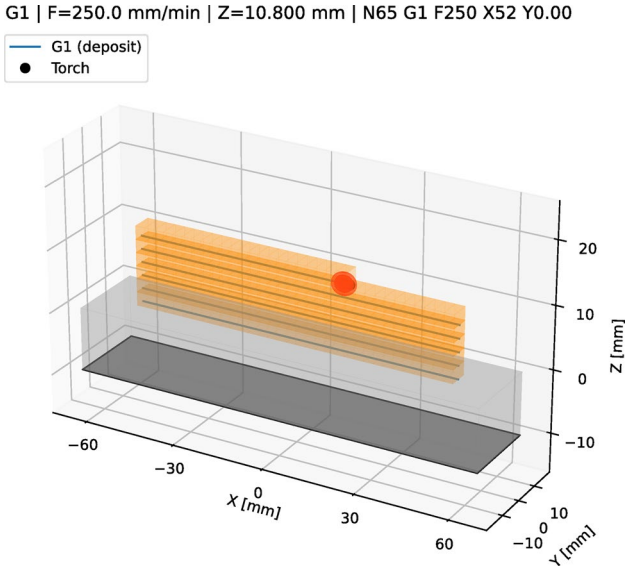


Fig. 5 Geometric reconstruction of the deposited thin wall based on G-code trajectories, including torch position and layer-by-layer material deposition

process states continuously update the active material domain, moving heat-source position, arc condition, and time-dependent boundary conditions. The workflow therefore separates the controller-specific instruction-parsing stage from the subsequent geometric and thermal calculations, allowing the latter to operate from a common process-state representation.

2.8 Thermal formulation and spatiotemporal discretization

2.8.1 Governing thermal equations

The thermal evolution of the WAAM process is described by the three-dimensional transient heat conduction equation with internal volumetric heat generation, since the temperature field varies simultaneously in the three spatial directions and in time within the solid domain formed by the deposited material and the substrate. This formulation is appropriate because heat is continuously introduced by the electric arc, while conduction governs its redistribution through the evolving wall and the base material, generating nonuniform thermal gradients and localized reheating as successive layers are added [23, 24]. The objective of this formulation is therefore to represent how the temperature field evolves in space and time during the deposition process.

The governing energy conservation equation is written as:

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{g} \quad (1)$$

where $T = T(x, y, z, t)$ is the temperature field, ρ, c_p, and k are the density, specific heat capacity, and thermal conductivity, and $\dot{g}$ is the volumetric heat generation associated with the electric arc. Assuming constant thermophysical properties, Eq. (1) reduces to the classical diffusion form:

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) + \frac{\dot{g}}{\rho c_p} \quad (2)$$

where the thermal diffusivity α is defined as:

$$\alpha = \frac{k}{\rho c_p} \quad (3)$$

The parameter α represents the rate at which heat propagates through the material, capturing the combined effect of heat conduction and thermal inertia.

Although WAAM involves large temperature variations, effective constant values of ρ, c_p, and k are adopted, consistent with simplified thermal modeling strategies previously applied in welding and WAAM to represent the dominant evolution of the thermal field [25–27]. From a numerical standpoint, this assumption reduces the complexity of the present explicit formulation by avoiding the repeated evaluation of temperature-dependent thermophysical properties. However, no claim is made regarding computational superiority relative to established finite-element or commercial thermal solvers [25, 26]. This choice is appropriate because the objective of the present study is not to reproduce the instantaneous peak temperature in the immediate vicinity of the arc, but to capture the dominant thermal response of the deposited wall, including cooling behavior, interlayer reheating, and heat accumulation, which are the quantities most relevant to the thermal–microstructural interpretation. In WAAM steels, the resulting microstructure is governed by repeated thermal cycling and cooling conditions along the build height rather than by the local maximum temperature at the heat source [25, 28]. Accordingly, the thermophysical properties are evaluated as representative average values over the temperature range most relevant to these transformations.

Within this scope, the present formulation should be regarded as a conduction-dominated thermal-history model rather than a melt-pool fluid-flow model. Temperature-dependent thermophysical properties, surface-tension gradients, and liquid-metal convection can influence melt-pool geometry and primary solidification conditions. In particular, Marangoni-driven convection requires explicit treatment of fluid flow and free-surface phenomena and therefore cannot be represented solely by introducing temperature-dependent ρ, c_p, or k into the present heat-conduction formulation. Consequently, the thermal–microstructural analysis

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performed here is not intended to predict primary solidification morphology directly from melt-pool physics. Instead, it uses the reconstructed post-deposition solid-state thermal histories to interpret differences in reheating frequency, interpass thermal state, cooling behavior, and exposure to transformation-relevant temperature ranges along the build height.

2.8.1.1 Boundary conditions Mixed convective–radiative boundary conditions are applied to all exposed surfaces of the voxelized domain:

$$-k\frac{\partial T}{\partial n} = h\left(T_s?T_\infty\right) + \epsilon\,\sigma\,\left(T_s^4?T_\infty^4\right) \tag{4}$$

where $\frac{\partial T}{\partial n}$ represents the temperature gradient normal to the surface, $h$is the convective heat transfer coefficient, $T_s$is the local surface temperature, T_∞ is the ambient temperature, ϵ is the emissivity, and σ is the Stefan–Boltzmann constant. Radiation cannot be neglected in WAAM because, while convection dominates at moderate temperatures, radiative losses become comparable or larger near the molten pool, and ignoring them would underestimate heat dissipation and overpredict thermal accumulation [23, 24, 27].

To reflect the directional cooling imposed experimentally, the convective coefficient is anisotropic and time-dependent. Surfaces exposed only to ambient air use a natural convection coefficient h_{nat}. The surface facing the auxiliary fan uses an enhanced coefficient h_{fan}, but only during the intervals in which the fan is active according to the G-code control signal; outside these intervals it reverts to h_{nat}. This synchronization between the convective boundary and the executed process signals is one of the elements that links the thermal model to the real machine behavior.

At the lower surface of the substrate, the experimental thermal insulation justifies an adiabatic condition:

$$\frac{\partial T}{\partial z} = 0 \tag{5}$$

neglecting heat losses toward the supporting structure while preserving the dominant heat-transfer mechanisms within the domain.

2.8.2 Moving Gaussian heat source

The volumetric heat input associated with the electric arc is modeled as a three-dimensional moving Gaussian distribution centered at the instantaneous torch position $(x_s(t), y_s(t), z_s(t))$, obtained directly from the G-code-driven trajectory [29–32].

$$\dot{g}(x,y,z,t) = \frac{P_{\text{eff}}}{\left(\sqrt{2\pi}\right)^3 \sigma_x \sigma_y \sigma_z} \exp\left[-\frac{1}{2}\left(\frac{(x - x_s(t))^2}{\sigma_x^2} + \frac{(y - y_s(t))^2}{\sigma_y^2} + \frac{(z - z_s(t))^2}{\sigma_z^2}\right)\right] \tag{6}$$

where the effective power input is defined as:

$$P_{\text{eff}} = \eta\, VI \tag{7}$$

with η being the process efficiency, V the arc voltage, and I the welding current.

The parameters σ_x, σ_y, and $\sigma_z$represent the standard deviations of the Gaussian distribution along each spatial direction and therefore control the spatial spread of the heat source. These parameters are directly related to the effective size of the molten pool and the distribution of thermal energy around the arc. In this way, the adopted formulation allows the heat input to be distributed smoothly around the instantaneous torch position, avoiding artificial discontinuities in the thermal field while preserving the localized nature of arc heating [29, 30, 32].

The heat-source parameters were selected independently of the thermographic comparison. The arc efficiency, $\eta = 0.85$, was adopted from values reported for welding-based thermal models [31, 32], while the Gaussian spread parameters were taken from the literature [32] as effective values representative of the spatial distribution of the arc heat input. The measured bead width and layer height were used to reconstruct the deposited geometry and define the progressive material-activation domain, rather than to calibrate the Gaussian heat-source parameters. No inverse calibration of the heat-source parameters against the thermographic measurements was performed. Accordingly, the adopted arc efficiency and Gaussian spread values should be regarded as effective modeling parameters, and variations in these quantities may influence the predicted peak temperatures and subsequent cooling behavior.

As illustrated in Fig. 6, the moving Gaussian heat source follows the deposition path defined by the G-code. As the heat source advances, new voxels are progressively activated, representing the addition of material, and therefore enabling the transfer of thermal energy into regions that were previously inactive. In this way, heat input is not applied to a fixed domain, but rather to an evolving geometry that grows layer by layer. Simultaneously, the set of boundary conditions is continuously updated as the domain expands. Newly activated external surfaces become subject to convective and radiative heat losses, while previously exposed surfaces may become internal as adjacent material is deposited. This dynamic coupling between heat input, progressive material activation, and time-dependent boundary conditions

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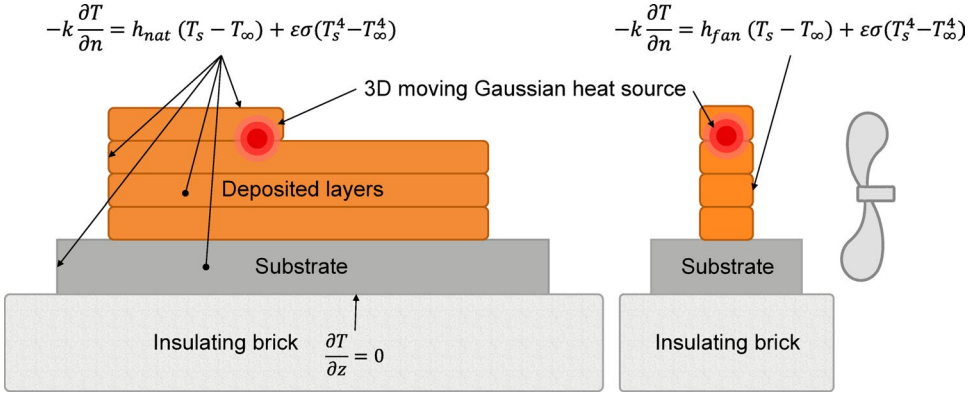


Fig. 6 Thermal boundary conditions in the WAAM model, including natural and forced convection, radiation, a moving Gaussian heat source, and an adiabatic base

is essential to accurately reproduce the transient thermal cycles characteristic of the WAAM process.

2.8.3 Numerical discretization

The transient thermal problem was solved over a structured voxel mesh using an explicit finite-difference time-marching formulation. The selection of this scheme was not based on an assumption that finite-element methods are unable to efficiently accommodate progressive material activation. Modern FEM formulations can represent sequential material addition through element birth-and-death or progressive activation strategies and offer important advantages in terms of stability and geometric flexibility. Instead, the present formulation was selected because it provides a direct correspondence between the structured voxelized geometry reconstructed from the G-code and the thermal computational domain.

Within the custom Python workflow, material activation, heat-source position, arc state, dwell periods, and time-dependent boundary conditions are synchronized directly with the machine-execution chronology. The structured-grid representation also provides direct programmatic access to the evolving temperature field and facilitates the customized extraction of local thermal-history descriptors. The choice should therefore be understood as an implementation and workflow decision tailored to the regular thin-wall geometry and process-driven objectives of the present study, rather than as a claim of general numerical or computational superiority over FEM.

A uniform cubic voxelization was adopted throughout the domain:

$$\Delta x = \Delta y = \Delta z = \Delta \tag{8}$$

where Δx, Δy, and Δz denote the spatial discretization along the Cartesian directions, and Δ is the uniform cubic voxel size.

This discretization provides a direct mapping between the reconstructed deposited geometry and the thermal grid, while simplifying the treatment of conductive fluxes and surface heat losses. Its main numerical limitation is the conditional stability of the explicit formulation, which couples the allowable time step to the spatial resolution and can substantially increase computational cost as the voxel size is refined. This limitation is explicitly evaluated through the mesh and time-step sensitivity analysis presented in Sect. 2.8.8.

2.8.4 Discretization of the governing equation in the active domain

Starting from the simplified governing equation already introduced in (2), the transient thermal field is discretized in time using a forward difference and in space using centered finite differences. This leads to the explicit update equation for an internal voxel:

$$\begin{aligned} T_{i,j,k}^{n+1} = & T_{i,j,k}^{n} + \Delta t \left[\alpha \left(\frac{T_{i+1,j,k}^{n} - 2T_{i,j,k}^{n} + T_{i-1,j,k}^{n}}{\Delta x^2} \right.\right. \\ & + \frac{T_{i,j+1,k}^{n} - 2T_{i,j,k}^{n} + T_{i,j-1,k}^{n}}{\Delta y^2} \\ & \left.\left. + \frac{T_{i,j,k+1}^{n} - 2T_{i,j,k}^{n} + T_{i,j,k-1}^{n}}{\Delta z^2} \right) \right. \\ & \left. + \frac{\dot{g}_{i,j,k}^{n}}{\rho c_p} \right] \end{aligned} \tag{9}$$

where i, j, and k denote the voxel indices in the x-, y-, and z-directions, respectively, and n denotes the time-step index.

The first term on the right-hand side is the temperature already stored in the voxel at time step n. The second term represents the thermal increment accumulated during the time interval Δt due to two contributions: heat diffusion from neighboring voxels and local volumetric heat input from the arc.

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The discrete second derivatives approximate the conductive exchange in the three coordinate directions. In physical terms, they measure the curvature of the temperature field and therefore determine whether heat is entering or leaving the voxel from its neighbors. The volumetric source term $\dot{g}^{n}_{i,j,k}$ accounts for the energy delivered by the moving Gaussian heat source at the voxel center during that same time step.

Although this discrete equation resembles the standard explicit formulation for transient conduction, in the present model it is applied only over the active portion of the domain, which makes the implementation fundamentally different from a classical fixed-domain thermal solver.

2.8.5 Conduction restricted to active voxels

A key aspect of the numerical implementation is that the computational domain evolves throughout the simulation rather than being assumed solid from the start. At the initial time, only the substrate is thermally active; deposited voxels are progressively incorporated as the torch advances along the G-code trajectory, so the active domain grows layer by layer in synchrony with the real deposition sequence.

Not-yet-deposited voxels are maintained at the ambient temperature, T_{∞} =293.15 K, while they remain thermally inactive. Upon activation, newly deposited voxels are initialized at T_{∞} and subsequently receive energy from the moving volumetric Gaussian heat source. This initialization represents a numerical convention of the present material-activation formulation rather than the physical temperature of the molten metal entering the weld pool. The thermal energy required to raise the temperature of the newly activated material is supplied through the effective volumetric heat-source term as the torch passes over the corresponding location. Once activated, the voxels participate in conductive heat transfer with neighboring active material and in the applicable surface heat-loss mechanisms.

Under this framework, conductive exchange is restricted to pairs of voxels that are both active at a given time step. Interfaces between an active voxel and a not-yet-deposited neighbor are excluded from the thermal solution, since allowing heat flow across them would produce a non-physical diffusion of energy into empty space and distort the predicted thermal history.

This progressive activation strategy preserves the coupling between deposition chronology and thermal evolution, which is particularly important in WAAM because each new layer is deposited onto a substrate whose temperature field has been shaped by all previous beads.

2.8.6 Discretization of the moving volumetric source term

The Gaussian source defined in (6) is evaluated at the center of each active voxel at every time step:

$$\dot{g}^{n}_{i,j,k} = \frac{P_{\text{eff}}}{(2\pi)^{3/2}\sigma_x\sigma_y\sigma_z} \exp\left[-\frac{1}{2}\left(\frac{(x_i - x_s(t_n))^2}{\sigma_x^2} + \frac{(y_j - y_s(t_n))^2}{\sigma_y^2} + \frac{(z_k - z_s(t_n))^2}{\sigma_z^2}\right)\right] \tag{10}$$

where (x_i, y_j, z_k) corresponds to the center of the voxel and $(x_s(t_n), y_s(t_n), z_s(t_n))$ is the instantaneous torch position obtained from the G-code trajectory. Since the voxel size is small compared with the Gaussian spatial spread, this pointwise evaluation provides a representative volumetric heat generation rate within each cell.

2.8.7 Discretization of boundary nodes

For voxels at the active-domain boundary, a control-volume energy balance is applied, incorporating conductive exchange with active neighbors, the local Gaussian source, and surface heat losses by convection and radiation. Equation (11) is written for the representative case of a voxel located on an exposed lateral wall of the deposit, having one free face and five active neighbors, which is the most frequent boundary configuration in the thin-wall geometry considered here:

$$\begin{aligned} kA\Bigg[&\frac{T^n_{i-1,j,k} - T^n_{i,j,k}}{\Delta x} + \frac{T^n_{i,j+1,k} - T^n_{i,j,k}}{\Delta y} + \frac{T^n_{i,j-1,k} - T^n_{i,j,k}}{\Delta y} \\ &+ \frac{T^n_{i,j,k+1} - T^n_{i,j,k}}{\Delta z} + \frac{T^n_{i,j,k-1} - T^n_{i,j,k}}{\Delta z}\Bigg] \\ &+ \dot{g}^n_{i,j,k}V - hA\left(T^n_{i,j,k} - T_{\infty}\right) - \varepsilon\sigma A\left[(T^n_{i,j,k})^4 - T^4_{\infty}\right] \\ &= \rho c_p V \frac{T^{n+1}_{i,j,k} - T^n_{i,j,k}}{\Delta t}. \end{aligned} \tag{11}$$

Here A=Δ² is the area of the exposed face and V=Δ³ is the voxel volume. Inactive neighbors are excluded from the conductive sum, so the same expression naturally adapts to the evolving free surface as deposition progresses. Edge and corner voxels are treated analogously, with the conductive terms of the missing neighbors removed and the convective and radiative terms applied over the two or three exposed faces, respectively. The convective coefficient is assigned according to face orientation and process state: h = h_{fan} for the face oriented toward the auxiliary fan while it is active according to the G-code control signal, and h = h_{nat} for all remaining exposed faces. At the lower substrate boundary, the convective and radiative terms vanish to enforce the adiabatic condition imposed by the experimental insulation arrangement, reducing the balance to conduction and internal generation only.

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2.8.8 Spatial and temporal resolution

The voxel size Δ and time step Δt are strongly coupled in an explicit scheme, and must be chosen as a compromise between geometric fidelity of the deposited bead, resolution of the thermal gradients around the moving source, and computational tractability over the full eight-bead deposition. Once Δ is fixed, the time step is constrained by the explicit-scheme stability requirement for three-dimensional diffusion:

$$Fo_x + Fo_y + Fo_z = \alpha \Delta t \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2} \right) \leq \frac{1}{2}. \quad (12)$$

Under the previously established assumption of a uniform cubic discretization, this condition simplifies to:

$$\Delta t \leq \frac{1}{2\alpha \left(\frac{3}{\Delta^2} \right)}. \quad (13)$$

A safety factor is applied throughout, $\Delta t = 0.3 \Delta t_{\max}$, to preserve robustness against radiative losses and time-dependent boundary conditions.

To identify an appropriate value of Δ, a mesh and time-step sensitivity study was performed on a single deposited bead followed by its cooling stage, which represents the fundamental thermal cycle repeated layer-by-layer during the full process, since refining the complete eight-bead sequence at the finest meshes was computationally prohibitive. Seven voxel sizes between Δ = 2.0 mm and Δ = 0.5 mm were tested, with Δt rescaled consistently through Eqs. (12) and (13). The transient temperature was tracked at a reference node centered in the substrate, directly below the bead.

As shown in Fig. 7, coarse meshes (Δ ≥ 1.5 mm) underestimate both the peak temperature and the cooling response, while the transient histories progressively converge as Δ decreases and essentially overlap for Δ ≤ 1.0 mm. Consistently, the final temperature at the reference node converges near 257 °C, whereas coarser meshes remain in the 220–230 °C range. The computational cost, however, scales approximately as Δ^{-5}, rising from ~4 s at Δ = 2.0 mm to ~50 min at Δ = 0.5 mm for a single bead, and is further amplified by the growing active domain in the full sequence.

Based on this trade-off, Δ = 1.0 mm is adopted as the reference resolution for the full-sequence simulations, lying within the converged regime while preserving the feasibility of simulating the complete eight-bead deposition.

2.9 Material properties and model parameters

The thermal model was parameterized using geometric, thermophysical, boundary-condition, process, and heat-source

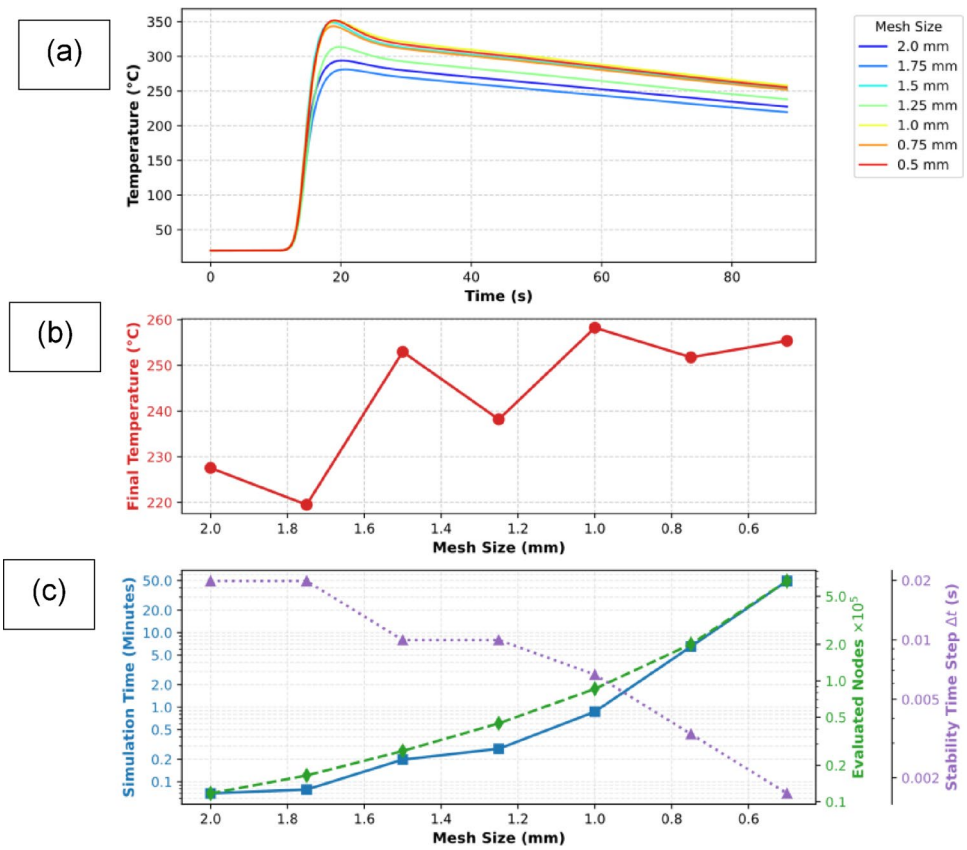


Fig. 7 Mesh and time-step sensitivity study on a single bead and its cooling stage: (**a**) transient temperature at the reference node for Δ = 0.5–2.0 mm, (**b**) final temperature as a function of Δ, and (**c**) associated computational cost in terms of simulation time, number of voxels, and stability-constrained Δt

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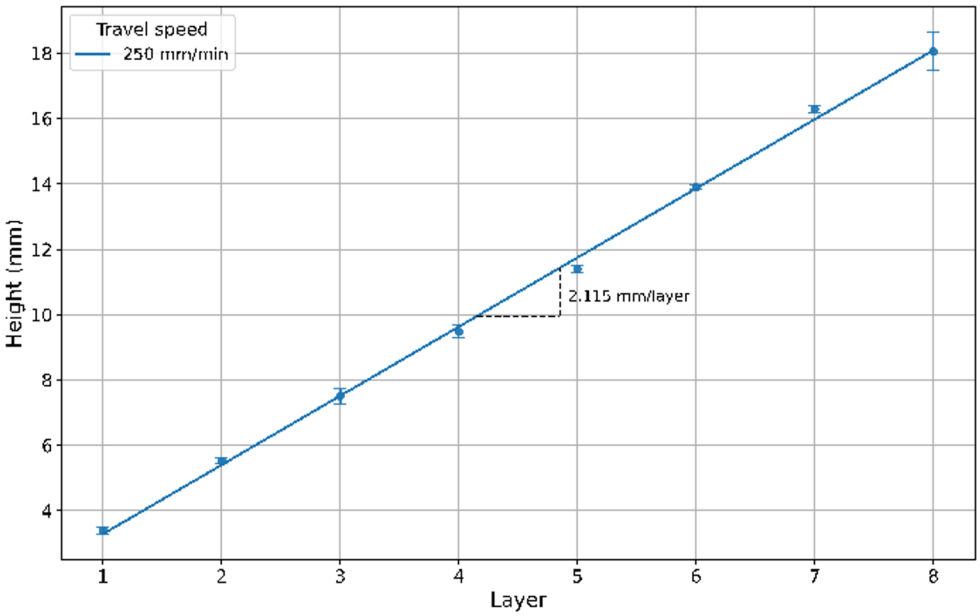


Fig. 8 Evolution of deposited wall height as a function of the number of deposited beads, showing a repeatable layer height of approximately 2.1 mm/layer and a final wall height close to 18 mm after eight layers

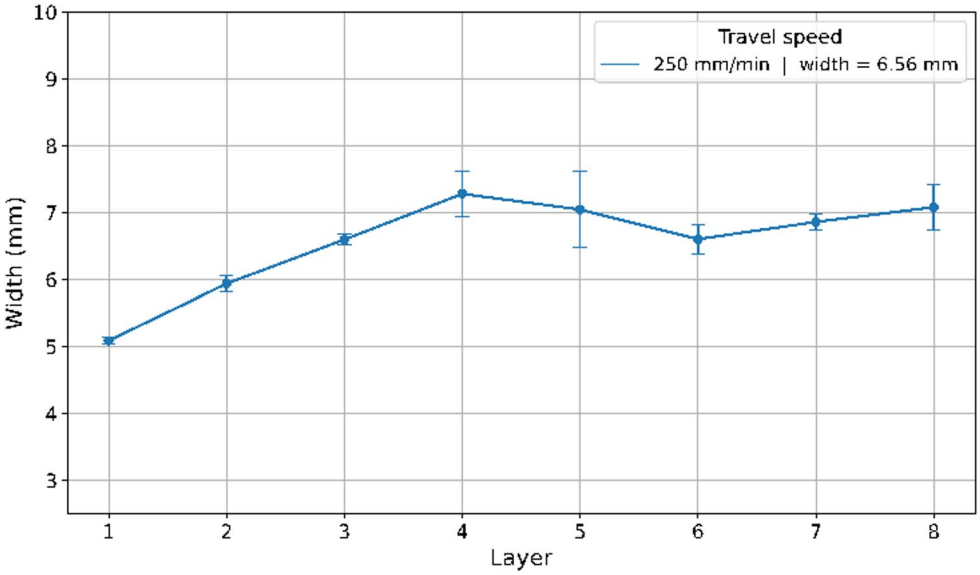


Fig. 9 Measured bead width for the deposited thin wall, showing an average value of 6.56 mm used as input for the geometric reconstruction in the thermal model

parameters consistent with the reference eight-bead thin-wall deposition. Instead of prescribing idealized bead dimensions, the reconstructed domain was based on measurements from the actual wall fabricated under the same processing condition used for thermographic comparison. As shown in Fig. 8, the layer height was determined as the slope of a linear fit of the measured wall height against the number of deposited beads, yielding a repeatable increment of approximately 2.1 mm per layer, with a final height close to 18 mm after eight layers. The first bead is taller than this steady-state increment, since it is deposited directly onto the cold substrate under different heat-extraction and dilution conditions, so the total height is not simply the product of the incremental layer height and the number of layers. The average bead width, shown in Fig. 9, was 6.56 mm. These values were used as direct inputs for the voxel-based geometric reconstruction and progressive material activation.

The thermophysical properties were adopted as effective average values representative of the intermediate-to-high solid-state temperature range relevant to microstructural evolution in ferritic-pearlitic steels. Particular emphasis was placed on the 500–800 °C interval, associated with the $\Delta t_{8/5}$ cooling range that governs austenite decomposition, diffusional ferritic transformations, and grain-growth phenomena in low-carbon steels processed by WAAM [16, 33]. Therefore, constant effective values were used to capture the dominant heat diffusion behavior relevant to the thermal–microstructural interpretation, rather than the full thermal excursion near the arc core. For naturally exposed surfaces, a convection coefficient of $h_{nat} = 10\, W/(m^2 \cdot K)$

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was adopted as a representative value for natural convection in air over solid surfaces [23].

For the wall face exposed to the auxiliary fan during interpass dwell periods, a forced-convection coefficient was estimated from the measured airflow velocity. Thermo-anemometer measurements indicated a nearly uniform air velocity of approximately 5 m/s over the cooled wall face. Because the airflow was directed approximately normal to the wall surface, the interaction was approximated as stagnation-type flow rather than a localized impinging jet. Accordingly, the forced-convection coefficient was estimated using the Falkner–Skan wedge-flow correlation for normal flow [34, 35]:

$$Re_L = \frac{UL}{\nu}, \qquad h_{fan} = \frac{Nu_L k_{air}}{L} \tag{14}$$

where Re_L is the Reynolds number based on the characteristic length L, Nu_L is the corresponding Nusselt number, U is the measured air velocity, ν is the kinematic viscosity of air, $k_{\text{air}}$is the thermal conductivity of air, and $h_{\text{fan}}$is the resulting forced-convection heat-transfer coefficient.

Using the measured air velocity and the adopted thermophysical properties of air, the Reynolds number was approximately 6494, yielding the corresponding Nusselt number and a forced-convection coefficient h_{fan} of $50.76\ W/(m^2 \cdot K)$. This value was applied uniformly to the active voxels on the cooled wall face during fan-on intervals, while the same surface returned to natural convection when the fan was inactive.

The electrical parameters used to define the moving heat source were taken from the monitored deposition condition. The arc voltage was 20 V, and the average measured current was 120 A. The remaining thermophysical, geometric, environmental, process, and numerical parameters used in the simulation are summarized in Table 2.

3 Results and discussion

3.1 Transient thermal field and qualitative thermographic comparison

The first step in the analysis of the numerical results is to compare the simulated temperature field with the thermal patterns observed experimentally during the WAAM process.

Figure 10 presents a qualitative comparison between the simulated thermal field mapped over the reconstructed 3D geometry and the corresponding thermographic views of the deposited wall at two selected time instants during fan-assisted cooling. The simulated fields are shown in the left column and the corresponding experimental thermographic images in the right column. The upper and lower rows correspond to approximately 404 s and 650 s, respectively, measured as elapsed time from the beginning of the experimental deposition sequence.

A clear similarity can be observed between the numerical and experimental thermal patterns. In both cases, the highest temperatures remain concentrated in the most recently deposited region, while lower temperatures extend progressively toward previously deposited portions of the wall and into the substrate. The results also show consistent qualitative agreement in the overall temperature range and in the longitudinal temperature gradient along the wall. Although the simulation and thermographic images are not expected to match pixel by pixel, the correspondence in the location of the hot region, the direction of thermal decay, and the extent of the heated zone indicates that the model captures the dominant mechanisms governing heat redistribution during interpass cooling, which is especially relevant because these snapshots correspond to fan-assisted cooling periods, where the thermal field is strongly influenced by both prior heat accumulation and directional convection. The fact that the model reproduces similar surface temperature distributions under these conditions supports the adequacy of the

Table 2 Material properties and model parameters used in the WAAM thermal simulation

Category	Parameter	Symbol	Value	Reference
Material	Density	ρ	7800 kg/m^3	[23]
Material	Specific heat capacity	C_p	800 J/(kg·K)	[23]
Material	Thermal conductivity	k	30 W/ (m·K)	[23]
Material	Surface emissivity	ϵ	0.7	[20, 21]
Environment	Ambient temperature	T_∞	293.15 K	Measured
Boundary condition	Natural convection coefficient	h_{nat}	10 W/ (m^2·K)	[23]
Boundary condition	Forced convection coefficient	h_{fan}	50.76 W/ (m^2·K)	[34, 35]
Process	Arc voltage	V	20 V	Measured
Process	Average measured current	$Imean$	120 A	Measured
Source model	Arc efficiency	η	0.85	[31, 32]
Source model	Gaussian spread in-plane	σ_{xy}	2 mm	[31]
Source model	Gaussian spread through thickness	σ_z	1.0 mm	[31]

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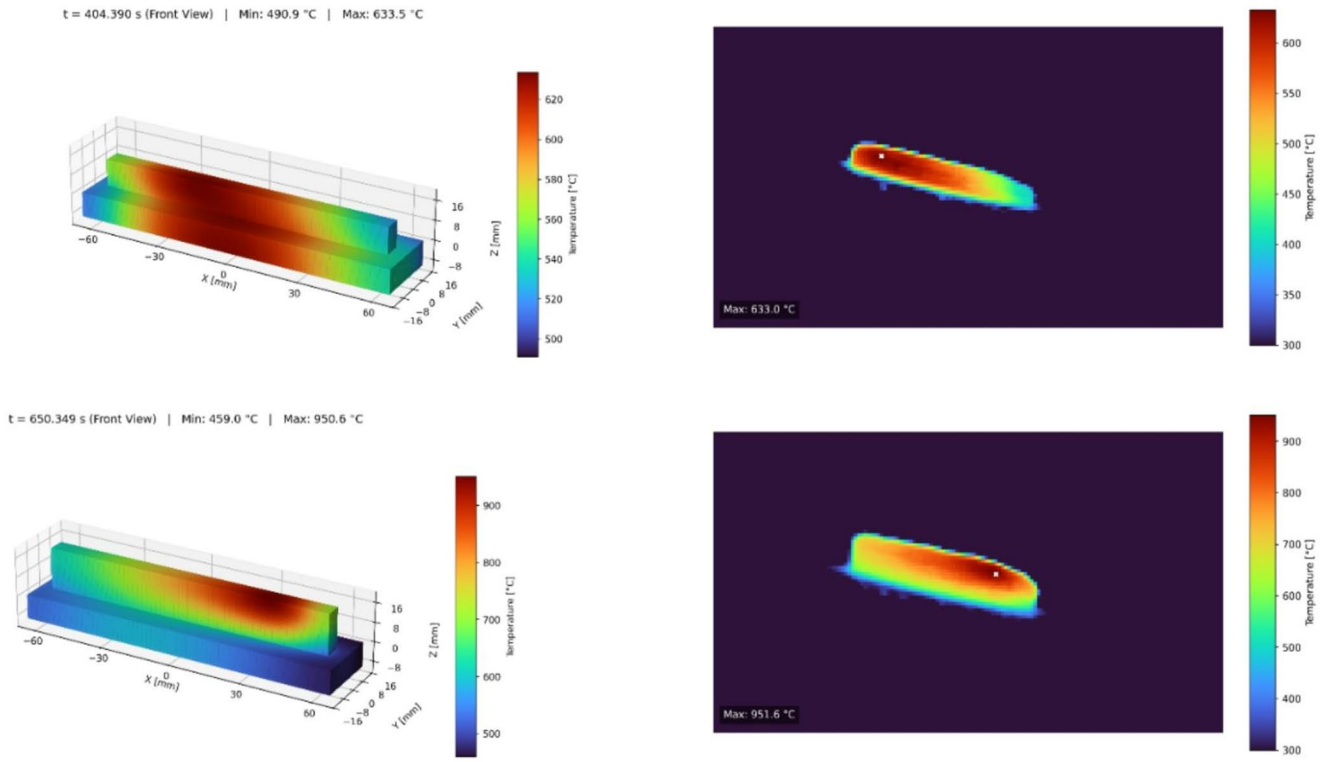


Fig. 10 Comparison between simulated 3D temperature fields (left column) and corresponding thermographic images (right column) at two equivalent time instants during fan-assisted cooling: approximately 404.39 s (top row) and 650.35 s (bottom row), measured from the beginning of the experimental deposition sequence

adopted conduction, convection, and radiation formulation for describing the cooling stage of the process.

Figure 11a shows the thermographic image at a time instant in which the torch is actively depositing, while Fig. 11b presents the corresponding simulated temperature field over the full numerical temperature range. In the experimental image, a clear saturation at high temperature is observed because the infrared camera is limited to a maximum measurable temperature of approximately 1200 °C. As a result, temperatures above this limit are clipped, preventing direct observation of the full thermal peak generated by the arc. In contrast, the numerical model is not affected by this measurement limitation and therefore predicts a wider temperature range, including temperatures above the thermographic saturation limit in the vicinity of the heat source.

However, when the simulated temperature field mapped over the 3D geometry is displayed using the same upper limit of 1200 °C as shown in Fig. 11c, the resulting thermal distribution becomes visibly much more similar to the thermographic observation. In this saturated representation, both the simulation and the experimental image exhibit comparable hot-zone shape, extent, and surface temperature gradient around the deposition front. This comparison indicates that the moving volumetric Gaussian heat source provides a physically meaningful representation of the actual heat input during deposition, since it reproduces not only the existence of a localized thermal peak but also the spatial spread of temperature around the torch position.

3.2 Quantitative comparison of thermal response

The quantitative comparison between the simulated thermal response and the infrared thermographic measurements was evaluated at both global and local scales. Because the infrared camera is limited to a maximum measurable temperature of approximately 1200 °C, the comparison was separated into saturation-affected regions and valid cooling intervals. Quantitative error metrics were calculated only over the latter, thereby avoiding the use of clipped thermographic measurements in the assessment.

Figure 12 compares the maximum surface temperature detected within the wall region observed by the thermal camera with the corresponding maximum simulated surface temperature throughout the complete deposition and interpass-cooling sequence. The upper panel shows the simulated and experimental temperature histories and identifies the saturation-affected intervals, whereas the lower panel presents the corresponding absolute and relative errors calculated only during the valid non-saturated cooling portions.

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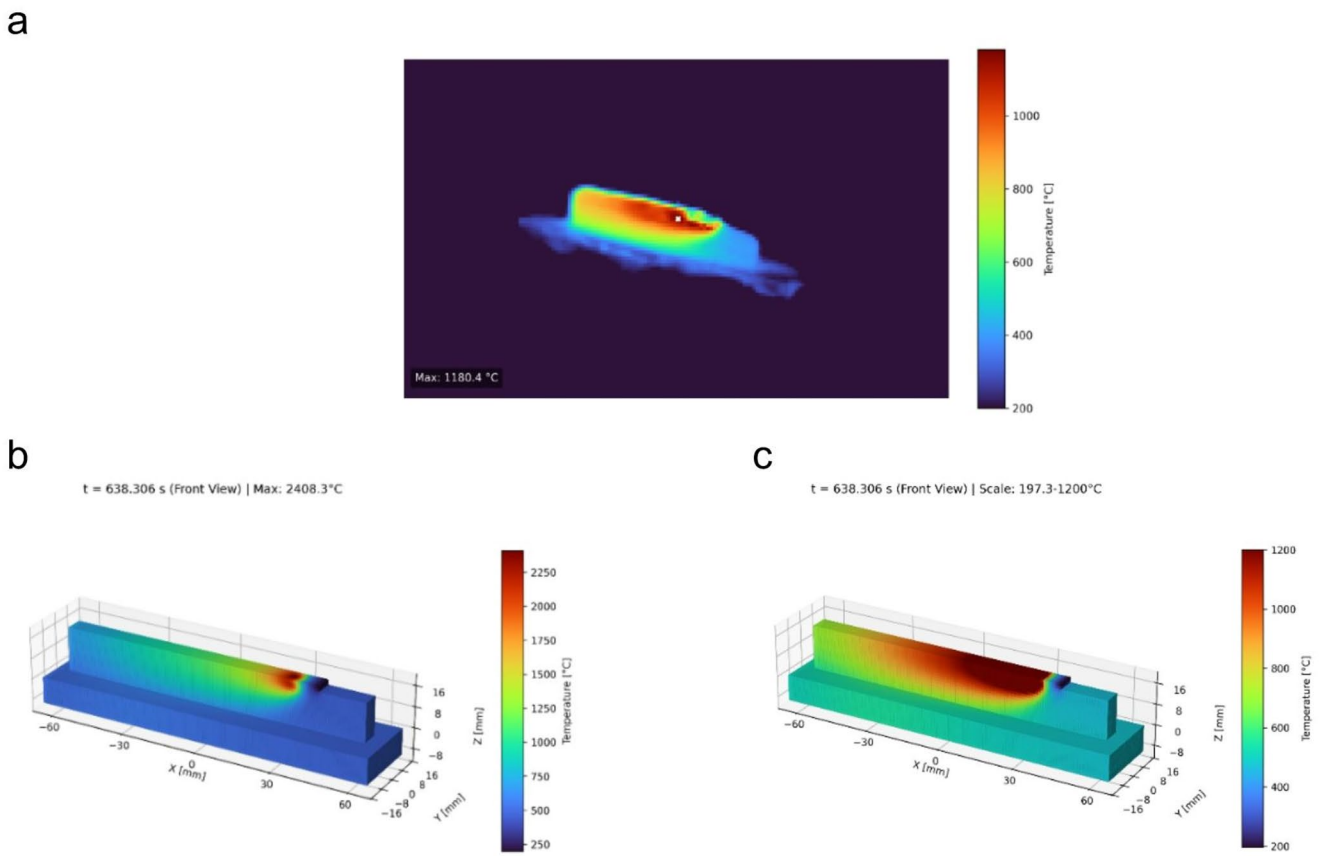


Fig. 11 Temperature field during active deposition: **a** thermographic image saturated at 1200 °C, **b** simulated field shown over the full thermal range, and **c** simulated field saturated at 1200 °C

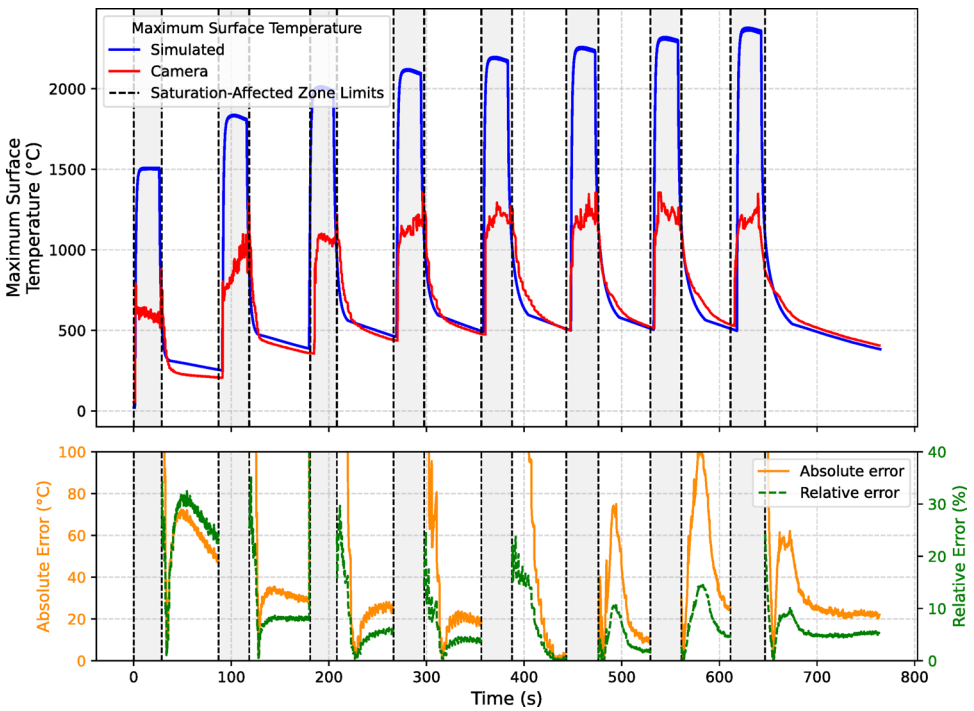


Fig. 12 Comparison of simulated and thermographically measured maximum surface temperatures during deposition and interpass cooling of the reference ER70S-6 WAAM specimen. The upper panel shows the simulated and experimental maximum surface-temperature histories, together with the saturation-affected interval limits, while the lower panel presents the corresponding absolute and relative errors calculated only over valid non-saturated cooling intervals

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The sequence of heating and cooling events is consistently reproduced by the simulation. During active deposition, the simulated surface temperature exceeds the measurement range of the camera, whereas the thermographic signal becomes clipped near its upper limit. Consequently, these intervals cannot be used for direct quantitative comparison. Once the heat source moves away and the measured temperature returns to the non-saturated range, both curves exhibit the characteristic rapid initial temperature decrease followed by a progressively slower cooling response. The corresponding error histories in Fig. 12 further show that the discrepancy is generally largest near the beginning of the valid cooling intervals and subsequently decreases as cooling progresses.

The largest differences during the initial stages are consistent with the strong conductive heat extraction imposed by the initially cold substrate. They are also consistent with the use of effective constant thermophysical properties in the numerical model. These properties were selected primarily to represent the intermediate-to-high temperature range relevant to the subsequent thermal–microstructural analysis and therefore do not reproduce the temperature dependence of carbon-steel properties over the entire thermal excursion. This limitation is particularly relevant during the first cooling stages, when a large portion of the thermal response rapidly enters the lower-temperature regime.

To complement this global comparison, Fig. 13 evaluates the thermal response at three representative surface locations along the build direction: Zone A near the first deposited bead and substrate, Zone B in the intermediate portion of the wall, and Zone C in the upper deposited region. For each location, the simulated history is compared with the thermographic signal extracted from the corresponding image position. The accompanying error plots show the absolute and relative discrepancies only during the valid portions of each cooling cycle.

The local histories reproduce the different thermal exposure experienced along the build height. Zone A undergoes its initial deposition followed by multiple reheating events as successive beads are added. Zone B becomes active later in the process and subsequently experiences several reheating cycles generated by the deposition of the upper layers. In contrast, Zone C is associated with the final deposited region and therefore experiences essentially its own deposition event followed by final cooling without further reheating. The simulated histories reproduce these distinct sequences of thermal peaks and cooling periods, while the quantitative error curves identify how the agreement evolves during each individual cycle.

The non-zero experimental signal observed in the intermediate and upper-region traces (regions B and C) before deposition of the corresponding beads does not represent

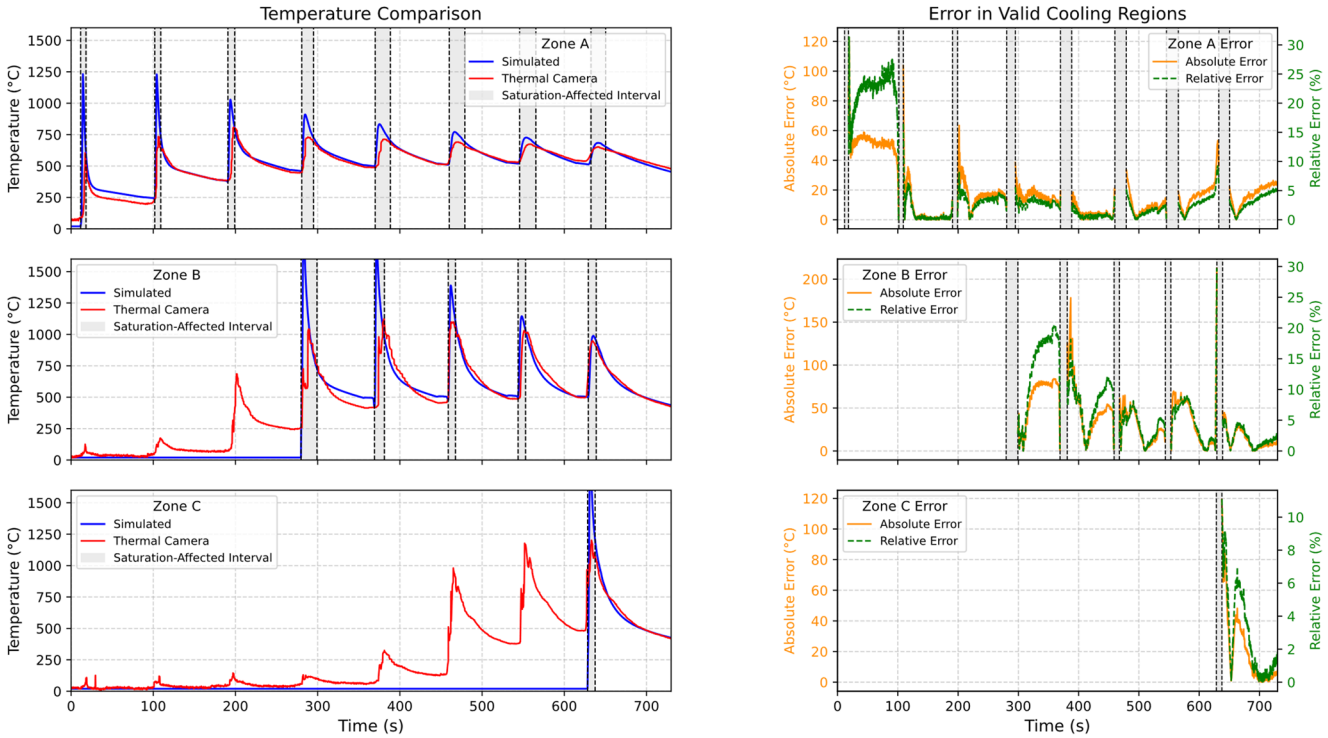


Fig. 13 Local comparison of simulated and thermographically measured temperature histories for three representative surface regions of the reference ER70S-6 WAAM specimen: Zone A (top row), Zone B (middle row), and Zone C (bottom row). The left column shows the simulated and experimental temperature histories together with the saturation-affected intervals, while the right column presents the corresponding absolute and relative errors calculated only over valid non-saturated cooling intervals

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the temperature of material that has not yet been deposited. Because the thermographic histories are extracted from projected image positions under an oblique viewing configuration and finite spatial resolution, the camera cannot resolve the thermal response at the same highly localized spatial scale as the numerical model. Consequently, the corresponding pixels or measurement areas can receive radiometric contributions from already-deposited underlying and neighboring hot surfaces, including the progressive heating of the surrounding region as the heat source approaches, even before material physically occupies the nominal measurement location. As the wall grows, the newly deposited material progressively becomes the dominant surface contributing to that projected measurement area. Therefore, the pre-deposition signals observed in regions B and C should be interpreted as a consequence of the finite spatial resolution and projected thermographic measurement geometry, rather than as the thermal history of not-yet-deposited material points.

The quantitative results for the individual cooling intervals are summarized in Table 3 using the signed bias, mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE). The largest discrepancy occurs during the first cooling event in Zone A, for which the MAE and RMSE are 52.0 °C and 52.8 °C, respectively, with a MAPE of 22.06%. After this initial event, the agreement improves substantially: the subsequent Zone A cycles exhibit MAPE values between 0.79 and 3.26%, with RMSE values between 6.1 and 19.9 °C.

A similar evolution is observed in Zone B. Its first valid cooling interval exhibits a comparatively larger discrepancy, with an MAE of 61.3 °C, an RMSE of 66.4 °C, and a MAPE of 13.42%. The relative discrepancy then decreases during the subsequent reheating cycles, with MAPE values of 7.97, 3.90, 4.87, and 2.10% for the thermal events associated with deposition of beads 5–8, respectively. Zone C, which is associated with the final deposited bead, exhibits an MAE of 20.7 °C, an RMSE of 31.6 °C, and a MAPE of 2.88% during its valid cooling interval. Overall, excluding the initial activation/cooling events of Zones A and B, all analyzed non-saturated intervals exhibit MAPE values below 8%, and most remain below 5%.

Table 3 Quantitative error metrics between simulated and thermographically measured temperatures for the valid non-saturated cooling intervals

Zone	Depositing bead	Bias (°C)	MAE (°C)	RMSE (°C)	MAPE (%)
A	1	51.9	52.0	52.8	22.06
A	2	−2.4	6.4	14.0	1.20
A	3	4.5	17.5	19.4	3.26
A	4	16.2	16.2	16.7	2.92
A	5	3.6	4.6	6.1	0.79
A	6	−2.2	9.3	11.2	1.60
A	7	−14.7	17.3	19.9	3.03
A	8	−13.9	16.7	18.0	3.13
B	4	57.0	61.3	66.4	13.42
B	5	−6.0	49.7	59.2	7.97
B	6	−11.0	26.2	31.2	3.90
B	7	−31.8	34.5	44.8	4.87
B	8	−0.6	13.1	16.9	2.10
C	8	−0.4	20.7	31.6	2.88

The bead number indicates the deposition event producing each heating/reheating cycle

The signed bias reported in Table 3 also shows that the model does not exhibit a uniform tendency to either overpredict or underpredict the measured temperature throughout the complete process. Positive and negative values occur depending on the location and thermal cycle, indicating that the discrepancy varies with the evolving thermal state rather than arising from a single systematic temperature offset.

These quantitative comparisons must nevertheless be interpreted considering the limitations of infrared thermography. Each experimental signal represents a finite projected surface area determined by the camera spatial resolution and oblique observation geometry. Consequently, the extracted temperature can contain contributions from adjacent regions having different thermal states, particularly in Zones B and C where strong local gradients coexist with the evolving wall geometry. In addition, uncertainty associated with the effective emissivity and the upper measurement limit of the camera restricts the comparison near the high-temperature deposition region. In contrast, the numerical model provides the temperature of a specifically defined active voxel and therefore has a substantially finer spatial definition than the experimental measurement.

Taken together, Figs. 12 and 13 and the quantitative metrics summarized in Table 3 demonstrate representative agreement between the simulated and thermographically measured thermal response for the selected WAAM condition. The agreement is strongest during the later reheating and cooling cycles, whereas larger discrepancies occur during the initial activation and cooling stages. Within these experimental and modeling limitations, the quantitative comparison supports the use of the simulated thermal histories for the subsequent interpretation of reheating frequency, cooling behavior, and microstructural variation along the build height.

3.3 Thermal–microstructural interpretation along the build height

Following the qualitative and quantitative comparison with the thermographic measurements, the simulated thermal histories are used to interpret the microstructural variations observed along the build height. The analysis focuses on a

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transverse cross-section of the thin wall fabricated under the same processing conditions represented in the simulation, allowing the experimentally observed microstructure to be spatially associated with the corresponding reconstructed thermal histories. Within the scope of the present conduction-based model, this interpretation is specifically concerned with post-deposition reheating, cooling behavior, and exposure to transformation-relevant temperature ranges. It therefore does not constitute a prediction of melt-pool flow, dendritic solidification, or primary nucleation mechanisms, which would require a higher-fidelity description of the molten-pool physics.

Figure 14 presents the reconstructed transverse section of the eight-bead wall together with the metallographic observations at 500× magnification, the corresponding grain-size segmentation maps, and the grain-size histograms for three representative regions labeled A, B, and C. In the segmentation maps, larger grains appear in warmer colors, particularly reddish and orange tones, whereas finer grains are represented by darker blue tones. The accompanying histograms quantify the area fraction associated with each grain-size bin, thereby providing a statistical description of the local microstructure rather than a purely qualitative visual impression.

A clear microstructural gradient is observed along the build height. The upper region, labeled C, corresponds to the most recently deposited material and therefore retains predominantly as-deposited characteristics because it is not subjected to subsequent interlayer reheating. Acicular ferrite remains visible within the prior-austenite grain domains, whereas coarser allotriomorphic ferrite develops preferentially along the former austenite grain boundaries and Widmanstätten side plates extend toward the grain interiors. These comparatively coarse and elongated ferritic features are reflected by the larger reddish regions in the segmentation map and by the broader grain-size distribution extending toward larger characteristic sizes. Such a morphology is consistent with the slower cooling conditions produced by progressive heat accumulation toward the upper part of the wall and with the absence of subsequent thermal cycles capable of substantially modifying the as-deposited structure [16, 33].

In contrast, the broad intermediate region, labeled B, exhibits a substantially more homogeneous microstructure dominated by equiaxed polygonal ferrite. The prior-austenite grain morphology and the coarse boundary-associated ferritic features observed in region C are no longer clearly preserved, while the segmentation map and corresponding histogram show a narrower and more uniform distribution centered at intermediate grain sizes. This difference is consistent with the repeated thermal cycling experienced by the intermediate layers. Successive reheating events, including repeated excursions into the austenitic temperature range, can partially or fully replace the previously formed ferritic structure through re-austenitization followed by subsequent transformation during cooling. Consequently, rather

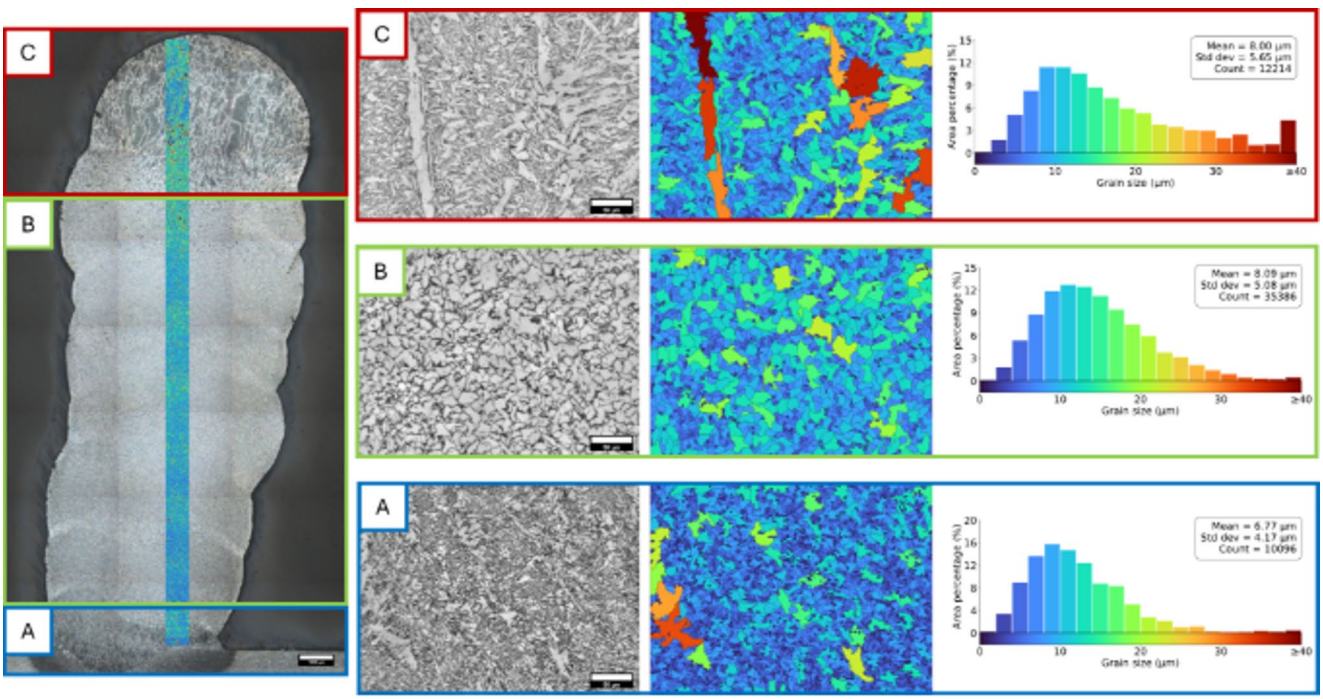


Fig. 14 Transverse-section reconstruction of the eight-bead WAAM wall and microstructural characterization of representative regions A, B, and C. For each region, the figure includes the metallographic image at 500×, the grain-size segmentation map, and the corresponding grain-size histogram

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than preserving the original as-deposited morphology, the repeated thermal history of region B is consistent with the progressive development of a more homogeneous equiaxed ferritic structure. Similar layer-dependent homogenization has been reported in previous studies of low-carbon steel WAAM [16, 33, 36, 37].

The lower region, labeled A, corresponding to the first deposited bead close to the substrate, shows a much finer microstructure. Here, the microstructure is characterized by very fine acicular ferrite, together with finer allotriomorphic ferrite and Widmanstätten side plates than those observed in the upper zone. As a result, the histogram shifts toward smaller grain sizes and the segmentation map is dominated by darker blue tones. From a thermal perspective, this refinement is associated with the more severe thermal extraction imposed by the substrate, which leads to a much faster cooling response immediately after deposition. Similar behavior has been reported in previous studies, where the lower layers were shown to contain finer acicular ferrite due to rapid cooling and stronger conductive heat extraction into the substrate [16, 33].

The role of epitaxial solidification also evolves as successive layers are deposited. During formation of the first bead, the molten ER70S-6 is in direct contact with the partially melted ASTM A36 substrate. Solid grains remaining at the fusion boundary can provide pre-existing crystallographic orientations from which solidification proceeds epitaxially, resulting in crystallographic continuity between the substrate and the initially solidified weld metal [38]. For subsequent beads, however, the fusion boundary is formed primarily with the previously deposited ER70S-6 rather than directly with the A36 substrate. Deposition of each new layer locally remelts the upper portion of the preceding bead, while solid material remaining below the fusion boundary can again provide crystallographic orientations for epitaxial growth. Consequently, epitaxial solidification may propagate progressively from layer to layer, whereas the direct metallurgical influence of the ASTM A36 substrate is strongest during formation of the first deposited bead. The resulting solidification structure is subsequently modified by the distinct reheating and cooling histories experienced at each build height.

Although a substantial part of the experimentally observed microstructural variation can be consistently interpreted from the reconstructed thermal histories, thermal cycling alone does not uniquely determine the final microstructure. Additional solidification- and melt-pool-scale mechanisms may contribute to the development of the primary structure and to the subsequent ferritic transformation. These include epitaxial nucleation and competitive growth from the substrate or previously deposited material at the fusion boundary, heterogeneous nucleation on non-metallic inclusions, and, under sufficiently undercooled conditions, mechanisms such as dendritic fragmentation or grain detachment that may introduce additional solid nuclei into the liquid [38–41]. In addition, the morphology and thermal state of the weld pool can be modified by fluid-flow phenomena driven by surface-tension gradients (Marangoni convection), buoyancy, and electromagnetic forces, which in turn affect local heat transport, fusion-zone geometry, temperature gradients, solidification rate, and therefore the resulting solidification structure [38]. Consequently, a modeling framework specifically focused on microstructural prediction would require the thermal-history description used here to be coupled with a higher-fidelity treatment of melt-pool fluid flow, free-surface behavior, solute redistribution, nucleation and growth kinetics, and phase-transformation mechanisms. The present model should therefore be interpreted as providing a physically consistent thermal basis for explaining the observed microstructural trends, rather than as a complete predictive model of microstructure formation.

Figure 15 complements this microstructural evidence by showing the simulated temperature histories for representative internal points located in the same cross-sectional region, together with the corresponding cooling paths plotted on a transformation diagram. The left side of the figure presents temperature as a function of time for sensors positioned in regions A, B, and C, while the right side shows the cooling trajectories associated with those regions whenever the local temperature exceeds Ac3, corresponding to full austenitization. In this way, the simulation provides a basis for identifying differences in reheating frequency and cooling behavior among the three regions. However, the predicted number and timing of Ac3 crossings should be interpreted as model-derived indicators rather than exact experimentally determined transformation events, since they may be sensitive to the adopted thermophysical properties, heat-source representation, boundary conditions, and emissivity. Therefore, the analysis emphasizes the relative thermal exposure of regions A, B, and C rather than the exact number of Ac3 crossings.

The sensor located in region A, corresponding to the first deposited bead, exhibits the fastest cooling behavior of all analyzed zones. Under the adopted model assumptions, this location is predicted to exceed Ac3 during its own deposition and once again after deposition of the second layer. Even after this second predicted excursion into the fully austenitic range, the subsequent cooling remains very rapid due to the strong conductive influence of the substrate. This thermal response is consistent with the experimentally observed fine acicular ferrite and the finer allotriomorphic and Widmanstätten products found in region A. Thus, the simulated history suggests that the lower bead remains strongly thermally influenced by the substrate and

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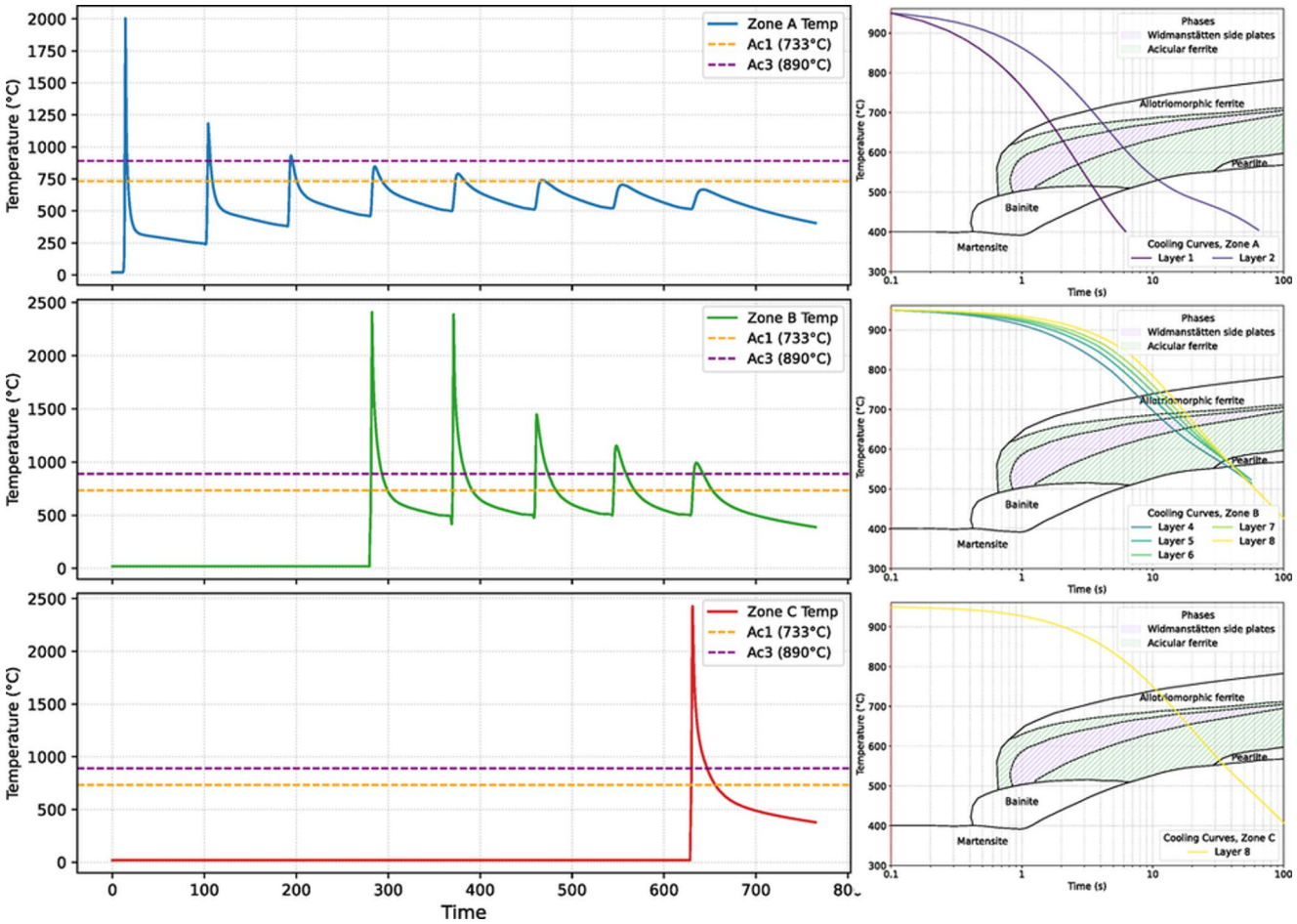


Fig. 15 Simulated thermal histories and transformation-based interpretation for representative regions A, B, and C of the wall cross-section. The left side shows temperature as a function of time for sensors located in each region, while the right side shows the corresponding cooling paths whenever the local temperature exceeds Ac3

experiences substantially less repeated reheating above Ac3 than the intermediate region, which is consistent with the finer microstructure experimentally observed in this region.

The sensor located in region B, approximately corresponding to the fourth bead, shows a markedly different thermal history. Under the adopted model assumptions, the simulated temperature in region B is predicted to exceed Ac3 five times, and the associated cooling cycles are generally slower than those recorded in region A. This higher reheating frequency and slower cooling provide a thermal basis for interpreting the substantial microstructural rearrangement observed experimentally, including the progressive suppression of initially finer ferritic products and the development of a more homogeneous equiaxed polygonal ferrite region. Rather than preserving the original as-deposited morphology, this intermediate region is consistent with repeated re-austenitization and transformation under progressively slower cooling conditions, favoring microstructural homogenization across the middle portion of the build.

Finally, under the adopted model assumptions, the sensor located in region C, corresponding to the eighth and last deposited bead, is predicted to exceed Ac3 only during its own deposition. Its cooling behavior is slower than that of region A and broadly comparable to the later cycles observed in region B, but unlike the intermediate region, it does not experience additional reheating from subsequently deposited material. This thermal history is consistent with the experimentally observed preservation of a predominantly as-deposited morphology in region C, consisting of acicular ferrite together with allotriomorphic ferrite and Widmanstätten side plates, but in a noticeably coarser form than in region A. The slower cooling is consistent with reduced refinement, while the absence of further reheating helps explain why this region does not exhibit the more homogenized equiaxed ferrite characteristic of the intermediate portion of the wall.

Taken together, the metallographic evidence and the simulated thermal histories show that the microstructure of the wall cannot be explained only by the last cooling cycle.

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Instead, it is controlled by the combined effect of cooling rate, reheating frequency, and whether or not a given region is repeatedly driven above Ac3. This is precisely where the simulation becomes essential. While the experimental micrographs reveal the final morphology, only the transient thermal model can reconstruct the full sequence of heating and cooling events that produced it. The simulation therefore provides a basis for interpreting how deposition strategy affects the final microstructure. This is particularly important in WAAM, where local thermal histories are position-dependent and strongly influence phase selection, grain refinement, microstructural homogenization, and mechanical response.

3.4 Model scope and limitations

The proposed framework was experimentally assessed under a single deposition condition and for one eight-bead thin-wall geometry. This configuration was deliberately selected as a controlled and repeatable case study, enabling direct comparison between the executed G-code, thermographic measurements, and metallographic observations. Therefore, the present results should be interpreted as a representative demonstration of agreement for the selected WAAM condition rather than as a comprehensive validation across multiple process regimes. Future work should evaluate the framework under different current–voltage combinations, travel speeds, interpass strategies, geometries, and materials.

Several numerical simplifications were also adopted. The use of effective constant thermophysical properties limits the accuracy of the model over the complete temperature excursion, particularly in the immediate vicinity of the heat source and during the earliest cooling stages. Temperature-dependent thermophysical properties can influence local heat transport, while melt-pool phenomena such as surface-tension-driven flow and Marangoni convection can affect fusion-zone geometry and primary solidification behavior. These effects are not explicitly represented in the present conduction-based formulation, whose primary objective is instead to reconstruct the subsequent reheating, cooling, and heat-accumulation histories of the deposited material. The moving volumetric Gaussian source therefore provides an effective representation of the arc heat input but does not explicitly resolve arc-plasma behavior, molten-pool convection, droplet transfer, or free-surface effects. Similarly, natural and forced convection are represented through effective heat-transfer coefficients, and the lower substrate boundary is approximated as adiabatic according to the experimental insulation strategy. Future extensions should incorporate temperature-dependent thermophysical properties and, where melt-pool-scale predictions are required, coupled fluid-flow or higher-fidelity thermal formulations.

The present framework should not be interpreted as a replacement for established finite-element platforms. In previous work by the authors on the same ER70S-6 WAAM system, transient thermal simulations performed in ANSYS using an element birth-and-death strategy successfully reproduced the principal layer-dependent thermal trends and supported the interpretation of microstructural evolution [16]. In that formulation, the deposited geometry and sequential element activation were defined within the finite-element preprocessing workflow before solution. The present voxel-based implementation was developed with a different objective: to provide direct programmatic coupling between the executed machine instructions and the evolving thermal model.

Both models reproduce the same principal physical trends observed experimentally, including rapid heat extraction near the substrate, repeated reheating of previously deposited material, progressive heat accumulation along the build height, and slower cooling in the upper region. This cross-method consistency provides complementary support for the ability of the present reduced thermal representation to reconstruct the dominant thermal evolution of the WAAM wall. However, because the two implementations were developed in separate studies under different process conditions and with different numerical and thermal settings, the previous FEM study is not considered a direct numerical benchmark of the present solver.

Accordingly, the main benefit of the current implementation is not claimed to be superior numerical accuracy or universal computational efficiency relative to commercial FEM software. Rather, its benefit lies in workflow flexibility: material activation, heat-source motion, dwell periods, auxiliary cooling states, and user-defined thermal-history outputs are generated directly from the processed machine instructions within the same Python environment. This reduces case-specific preprocessing when the deposition sequence is modified and facilitates customized extraction of local thermal descriptors. A controlled one-to-one comparison with an established FEM implementation under identical thermal, geometric, and process conditions remains an important direction for future work.

Finally, the uniform Cartesian voxelization introduces an inherent geometric limitation. Curved, inclined, non-orthogonal, or freeform boundaries are represented through discrete voxel faces and may therefore exhibit stair-stepping errors, whose magnitude decreases with spatial refinement at the expense of increased computational cost. The present eight-bead thin-wall geometry is comparatively favorable for this representation because the deposition path and principal wall surfaces are predominantly aligned with the

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Cartesian grid, while the experimentally measured bead width and layer height are used to define the activated domain. Nevertheless, the voxelized geometry remains an approximation of the actual deposited surface. For more geometrically complex WAAM components, conformal finite-element discretizations may provide greater geometric fidelity. Future work should therefore investigate the transfer of the same machine-execution-driven process representation to FEM-based domains and compare both formulations under equivalent conditions.

4 Conclusions

A three-dimensional transient thermal framework for WAAM was developed by integrating established thermal-modeling components within a machine-execution-driven computational workflow. The contribution does not lie in the individual use of progressive voxel activation, a moving Gaussian heat source, or transient heat conduction, which are established approaches in additive-manufacturing simulation. Instead, the framework uses the same post-processed CNC instructions employed during fabrication to reconstruct the time-resolved sequence of torch motion, arc on/off states, non-deposition movements, interpass dwell periods, and fan-assisted cooling events. These process states synchronously control material activation, heat-source motion, and time-dependent boundary conditions, while experimentally measured bead dimensions inform the deposited geometry. The resulting architecture provides a process-consistent reconstruction of the thermal history at the execution level of the investigated CNC-based WAAM system.

The model demonstrated representative agreement with thermographic measurements for the selected eight-bead WAAM condition. Simulated temperature fields agreed with infrared thermographic observations both qualitatively, in the spatial distribution and direction of thermal decay during active deposition and fan-assisted cooling, and quantitatively, in the global evolution of the hottest surface region and in the local histories of selected points along the wall. This agreement should be interpreted within the saturation, spatial-resolution, oblique-viewing, and emissivity-related uncertainties inherent to the thermographic measurements.

The simulated thermal histories showed that the local thermal response in WAAM is strongly position-dependent and cannot be described by a single representative cycle. Under the adopted model assumptions, the first deposited bead is predicted to cross Ac3 twice and to cool rapidly under the strong thermal influence of the substrate, whereas the intermediate region is predicted to experience five Ac3-crossing reheating events under progressively slower cooling conditions. The upper region, in contrast, is predicted to exceed Ac3 only during its own deposition and to undergo no subsequent reheating from additional layers. Although the exact number and timing of these threshold crossings are model-dependent, the relative differences in reheating frequency and cooling behavior are consistent with the observed microstructural gradient: fine acicular ferrite near the substrate, homogeneous equiaxed polygonal ferrite in the intermediate region, and a coarser as-deposited ferritic morphology in the upper layers.

Beyond reproducing temperature fields, the framework shows that the experimentally observed microstructural variation along the build height is consistent with differences in cooling rate, reheating frequency, and repeated excursions above Ac3 reconstructed from the executed deposition sequence. This establishes a quantitative basis for using the G-code itself as a design variable for process parameters, cooling strategies, and path planning in WAAM of low-carbon steels.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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